

FPGA Based Ultrasonic Radar System using AT-STLN-ARTIX 7-001

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Abstract

This project presents the design and implementation of a real-time ultrasonic radar system built around a Xilinx Artix-7 Field Programmable Gate Array (FPGA). The system integrates an HC-SR04 ultrasonic distance sensor mounted on an SG90 micro servo motor, both of which are directly controlled by custom Verilog modules running on the FPGA fabric. The FPGA generates the pulse-width modulation (PWM) signal required to sweep the servo across a defined angular range, triggers the ultrasonic sensor at each angular position, and measures the echo pulse width using a high-resolution hardware counter to compute object distance.

The computed angle and distance pairs are packaged and transmitted over a UART serial link at 115200 baud to a host PC, where a Processing 4 sketch parses the incoming data stream and renders a live, sweeping radar display, complete with a fading trail and highlighted target markers for detected objects. The complete signal chain, from servo actuation and echo timing to serial transmission and visualization, is implemented with deterministic, parallel hardware logic rather than sequential microcontroller code, demonstrating the advantages of FPGA-based design for real-time embedded sensing applications.

The prototype was implemented on a custom Artix-7 development board, and testing confirmed smooth servo sweeping, accurate short-range distance measurement, reliable UART transmission and correctly synchronized radar visualization. The project demonstrates core competencies in digital design, sensor interfacing, timing-critical hardware description and host-side data visualization, and lays the groundwork for further enhancements such as 360-degree scanning, wireless telemetry and object tracking.

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Chapter 1: Introduction

Radar (Radio Detection and Ranging) is a sensing technique used to detect the presence, distance and bearing of objects by analyzing reflected energy. While conventional radar systems rely on electromagnetic waves, the same fundamental sweep and detect principle can be replicated at low cost using ultrasonic sound waves, giving rise to compact “ultrasonic radar” demonstrators that are widely used in robotics and embedded systems education.

This project implements such an ultrasonic radar system using an HC-SR04 ultrasonic distance sensor mechanically mounted on an SG90 micro servo motor. The servo sweeps the sensor across an angular field of view, and at each angular step the sensor measures the distance to the nearest reflecting object. Unlike typical implementations that rely on a microcontroller such as an Arduino, this project uses a Xilinx Artix-7 FPGA as the central controller, with all timing critical operations, PWM generation, echo measurement and UART transmission, implemented directly in Verilog HDL.

1.1 Motivation

Microcontroller based radar projects are common, but they process tasks sequentially and are subject to interrupt latency and instruction execution jitter when handling multiple simultaneous timing critical operations. An FPGA, in contrast, executes independent hardware modules in true parallelism. This project was motivated by the opportunity to explore these advantages directly: generating servo PWM, measuring sub-microsecond echo pulse widths and streaming UART data all run as concurrent, independent hardware processes on the same chip, with no operation blocking another.

1.2 Objectives

- To design and implement FPGA-based digital hardware for interfacing with an HC-SR04 ultrasonic sensor and an SG90 servo motor.
- To generate accurate PWM control signals in hardware for continuous angular sweeping of the servo.
- To implement a precise echo pulse-width measurement module for ultrasonic distance computation.
- To design a UART transmitter module capable of streaming angle-distance data to a host PC at 115200 baud.
- To develop a Processing 4 visualization application that renders the incoming serial data as a live, animated radar sweep.
- To verify end-to-end functionality through hardware testing, covering servo motion, distance accuracy, UART integrity and visualization synchronization.

1.3 Scope of the Project

The scope of this mini project is limited to a single sensor, 2D angular sweep radar covering a forward arc (typically 0° to 180°), implemented on a single FPGA board with wired UART connectivity to a host PC for visualization. Extensions such as full 360-degree coverage, wireless telemetry and multi-target tracking are identified as future scope rather than part of the present implementation.

Chapter 2: Literature Survey

This chapter reviews the foundational concepts and technologies that underpin the project: radar principles, ultrasonic sensing, FPGA architecture, the Verilog hardware description language, servo motor control and UART-based serial communication, along with the Processing visualization environment.

2.1 Radar Systems

Radar systems operate by emitting a directed pulse of energy and measuring the time taken for the reflected echo to return from a target. The round trip time, combined with the known propagation speed of the energy through the medium, yields the distance to the target. Mechanically rotating or sweeping radar antennas extend this single-direction ranging into a 2D map of the surrounding environment, a principle directly mirrored in this project's sweeping ultrasonic sensor.

2.2 Ultrasonic Sensing and the HC-SR04

The HC-SR04 module operates by emitting an ultrasonic burst at approximately 40 kHz from a transmitting transducer upon receiving a trigger pulse, and measuring the time until the reflected echo is detected by a receiving transducer. The distance is computed from the round trip time of flight using the speed of sound in air. Ultrasonic sensors are favored in low cost embedded projects for their simplicity, low power consumption and adequate accuracy for short-range obstacle detection.

2.3 FPGA Technology

Field Programmable Gate Arrays are reconfigurable digital integrated circuits composed of arrays of programmable logic blocks, interconnect routing and dedicated resources such as block RAM and DSP slices. Unlike microcontrollers, which execute instructions sequentially on a fixed processor core, FPGAs are configured to implement custom digital circuits where independent logic blocks operate concurrently. This parallelism enables deterministic, low latency timing behavior that is well suited to applications such as PWM generation and precision pulse measurement, both of which are central to this project.

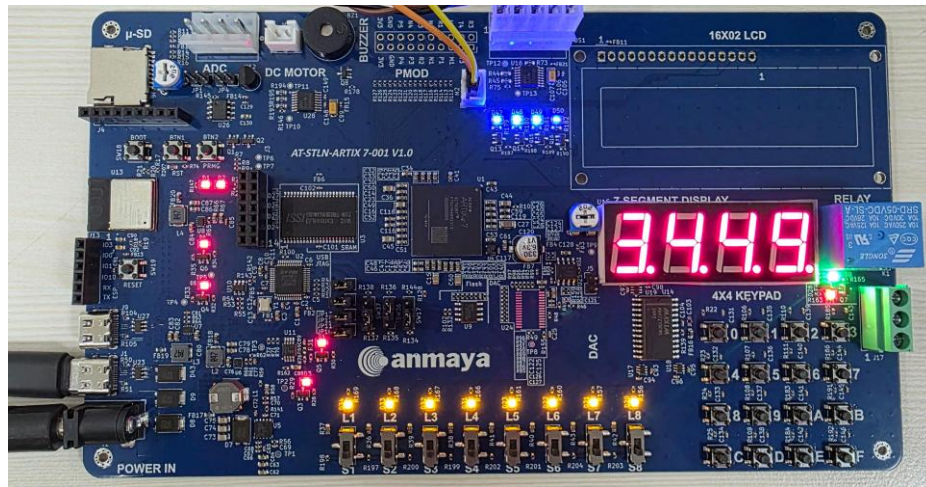


Figure 2.1: Custom AT-STLN Artix-7 FPGA Development Board Used in the Project

2.4 Verilog

Verilog is a hardware description language used to model digital systems at the behavioral and structural level for synthesis onto FPGAs and ASICs. Designs are described as interconnected modules, each representing a piece of hardware, and synthesis tools such as Xilinx Vivado translate the HDL description into a configuration bitstream for the target device. This project's Verilog modules describe the servo controller, ultrasonic interface, UART transmitter and top-level integration logic.

2.5 Servo Motor Control

The servo motor SG90 is positioned using a PWM control signal with a nominal period of 20 ms, where the pulse width (typically between 1 ms and 2 ms) determines the commanded shaft angle. Generating this signal in FPGA hardware, rather than via software delay loops, ensures jitter-free, continuously updating servo positioning independent of other system activity.

2.6 UART Communication

Universal Asynchronous Receiver-Transmitter (UART) communication is a simple, widely supported serial protocol that transmits data as a sequence of start bit, data bits, optional parity bit and stop bit(s), with both ends of the link agreeing on a common baud rate in advance. UART was selected for this project due to its simplicity of hardware implementation and broad compatibility with PC serial terminals and the Processing environment via a USB-to-serial (FTDI) bridge.

2.7 Processing Visualization Environment

Processing is a Java-based programming environment widely used for visual and interactive applications. Its built-in serial communication library allows it to read live

data streams from microcontrollers or, as in this project, an FPGA-driven UART link, and its 2D graphics primitives are well suited to rendering animated radar-style sweep displays.

2.8 Summary of Literature Survey

The reviewed literature confirms that FPGA-based implementations offer deterministic timing and true hardware parallelism that are advantageous for multi-peripheral, timing-critical applications such as this radar system, when compared to the sequential execution model of conventional microcontrollers.

Chapter 3: Proposed System

3.1 System Overview

The proposed system consists of an Artix-7 FPGA acting as the central controller for an HC-SR04 ultrasonic sensor mounted on an SG90 servo motor. The FPGA continuously sweeps the servo across its angular range, triggers a distance measurement at each step, computes the corresponding distance, and transmits the angle-distance pair over UART to a host PC, where a Processing 4 application renders the result as a live radar display.

3.2 Hardware Requirements

Component	Description / Role
Artix-7 FPGA Board	Central controller; hosts all Verilog logic for PWM generation, echo timing and UART transmission
HC-SR04 Ultrasonic Sensor	Emits ultrasonic pulses and detects echo for distance measurement
SG90 Micro Servo Motor	Sweeps the ultrasonic sensor across the angular field of view
FTDI USB-to-UART Module	Bridges the FPGA's UART TX line to the host PC's USB serial port
Component	Description / Role
5V Regulated Power Supply	Powers the servo motor and sensor module

Table 3.1: Hardware components used in the proposed system

3.3 Software Requirements

Tool	Purpose
Xilinx Vivado Design Suite	Verilog synthesis, implementation, bitstream generation and FPGA programming
Processing 4	Host-side serial data visualization (radar sweep display)
PuTTY	Raw UART terminal access for debugging serial output

Table 3.2: Software tools used in the proposed system

3.4 System Operation

The overall operation of the system proceeds through the following sequence at each angular step of the sweep:

1. The servo controller module increments or decrements the commanded angle and updates the PWM duty cycle accordingly, sweeping the sensor to the next angular position.

2. After allowing time for the servo to settle, the ultrasonic sensor module issues a 10 μ s trigger pulse to the HC-SR04.
3. The module measures the width of the resulting echo pulse using a hardware counter clocked by the FPGA system clock.
4. The measured echo width is converted into a distance value using the speed of sound.
5. The current angle and computed distance are formatted into a comma separated ASCII string and transmitted via the UART TX module at 115200 baud.
6. The Processing application receives the data over the serial port, parses the angle and distance fields, and updates the radar display, sweeping a beam line and plotting a marker at any detected target.

3.5 System Block Diagram

The signal flow of the system is summarized below: the top-level FPGA design coordinates the servo controller and ultrasonic sensor modules, both of which feed into the UART transmission path that ultimately drives the Processing-based radar display on the host PC.

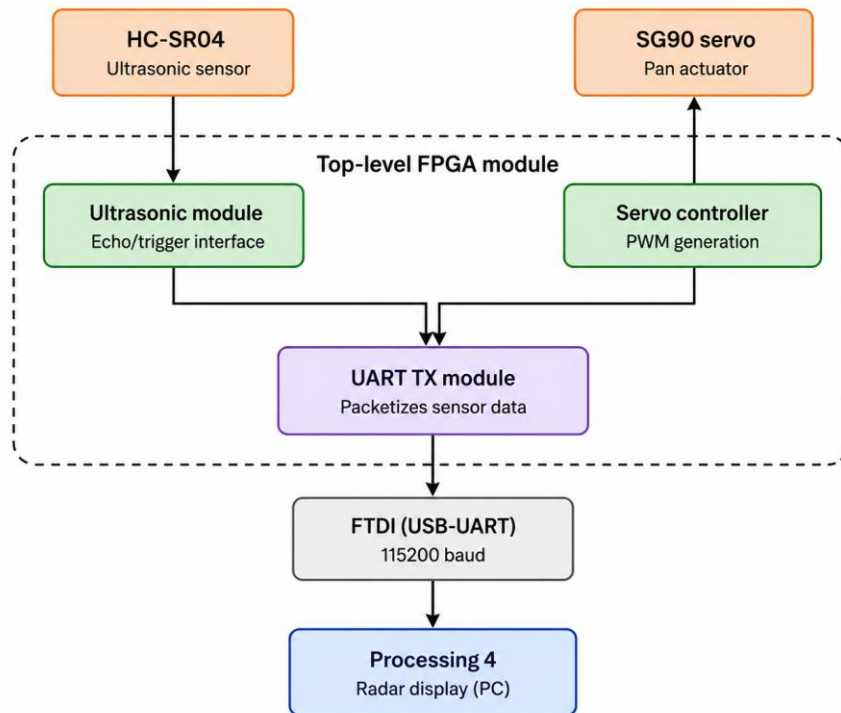


Figure 3.1: High-level signal-flow block diagram of the proposed system

Chapter 4: Implementation

The FPGA design is partitioned into five Verilog modules, each responsible for a distinct, independently operating function. This modular structure mirrors standard FPGA design practice, isolating timing-critical logic (PWM generation, echo measurement, serial transmission) into separate, reusable, and individually verifiable blocks.

4.1 Module Hierarchy

Module	Function
Top	Top-level module instantiating and interconnecting all sub-modules; distributes clock and reset
Servo Controller	Generates the 20 ms period PWM signal for the SG90 and sweeps the commanded angle
Ultrasonic Sensor	Generates the HC-SR04 trigger pulse and measures echo pulse width via hardware counter
UART TX	Implements the UART transmit shift register and baud-rate generator at 115200 baud
Radar UART	Formats angle and distance values into the “AAA,DDD” ASCII packet and feeds the UART TX module

Table 4.1: FPGA module hierarchy and responsibilities

4.2 Servo Controller Module

The servo controller module produces a PWM waveform with a fixed period of 20 ms, where the high-time of the pulse, typically ranging between approximately 1 ms (0°) and 2 ms (180°), is derived from a counter that increments the target angle at a controlled rate. This produces a smooth, continuous sweeping motion of the servo without requiring any software intervention, since the entire waveform is generated by a free-running hardware counter compared against the current duty-cycle register.

4.3 Ultrasonic Sensor Module

This module issues a 10 μ s trigger pulse to the HC-SR04 at the start of each measurement cycle. It then monitors the sensor's echo line and uses a high frequency hardware counter to measure the duration for which the echo signal remains high. Since the FPGA clock provides a fine time resolution, this counter value can be converted into a precise distance measurement using the known speed of sound, without the timing uncertainty that a software-based pulse measurement on a microcontroller would introduce due to interrupt latency.

4.4 UART TX Module

The UART transmitter module implements a standard asynchronous serial transmit path: a baud-rate generator derives a 115200 baud bit-clock from the FPGA system

clock, and a shift register serializes each outgoing byte, framed with a start bit and stop bit, onto the UART TX line. The module exposes a simple data/ready handshake to the Radar UART module so that bytes can be queued for transmission as soon as the line is free.

4.5 Radar UART Module

The Radar UART module is responsible for packaging the current angle and measured distance into a fixed ASCII format and feeding it byte-by-byte to the UART TX module. The chosen packet format is “AAA,DDD”, where AAA represents the three-digit servo angle in degrees and DDD represents the three digit measured distance in centimeters, separated by a comma and terminated appropriately for parsing on the host side.

4.5.1 UART Packet Format

Field	Width	Description
AAA	3 digits	Current servo sweep angle in degrees (000–180)
','	1 char	Field separator
DDD	3 digits	Measured distance in centimeters

Table 4.2: UART packet structure (115200 baud, 8-N-1)

4.6 Top-Level Integration

The top-level module instantiates the servo controller, ultrasonic sensor, UART TX and Radar UART modules, connecting them through a shared system clock and reset network. The servo controller's current angle output is shared with the Radar UART module so that each transmitted packet correctly reflects the sensor's instantaneous sweep position at the time of measurement, ensuring that angle and distance data remain synchronized in the resulting visualization.

4.7 Host-Side Visualization (Processing 4)

On the host PC, a Processing 4 sketch opens the serial port exposed by the FTDI bridge at 115200 baud and continuously reads incoming “AAA,DDD” packets. Each received packet is parsed to extract the angle and distance fields, which are then used to render a sweeping radar beam line at the corresponding angle. When a valid target is detected within sensor range, a red marker dot is plotted at the corresponding angle and radial distance on the circular radar display, with a fading trail effect applied to recently swept angles to emulate a traditional radar scope.

4.8 Tools and Design Flow

The Verilog source files were developed and synthesized using Xilinx Vivado, targeting the Artix-7 device. The design flow followed standard FPGA practice: RTL design and simulation, synthesis, implementation (placement and routing), and bitstream generation, followed by programming the device and validating module behavior using

onboard LEDs and a PuTTY serial terminal before connecting the Processing visualization.

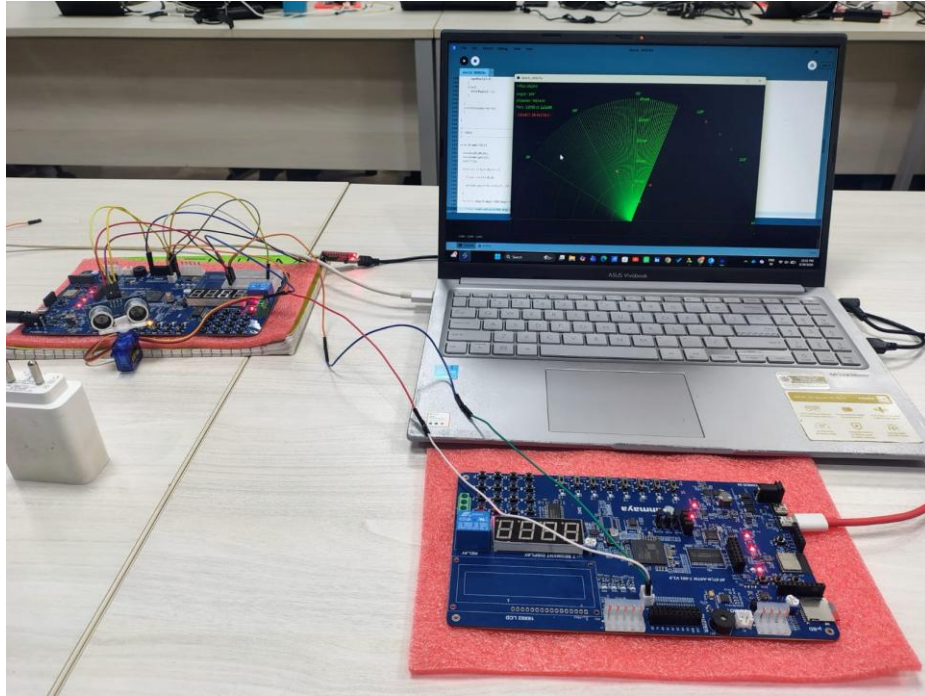


Figure 4.1: Experimental Hardware Setup of the FPGA-Based Ultrasonic Radar System

Chapter 5: Results

The completed system was tested on the target Artix-7 hardware platform, with the HC-SR04 and SG90 servo connected via the board's GPIO headers and the UART TX line routed through an FTDI module to the host PC.

5.1 Functional Testing

Test	Observation	Result
Servo sweep motion	Smooth, continuous sweeping across the configured angular range with no visible jitter	Pass
Echo pulse measurement	Consistent, repeatable distance readings for objects placed at known distances	Pass
UART data transmission	Correctly framed "AAA,DDD" packets observed in PuTTY at 115200 baud	Pass
Radar visualization sync	Processing display sweep tracked the physical servo position with no perceptible lag	Pass

Table 5.1: Summary of functional test results

5.2 Discussion

The results confirm that the FPGA-based architecture successfully performs real-time servo sweeping, ultrasonic distance measurement, UART transmission and host-side radar visualization in a fully synchronized manner. The hardware parallel implementation of the PWM and echo-timing modules ensured that the servo motion and distance measurement cycles did not interfere with one another, demonstrating the timing advantages of an FPGA-based approach over an equivalent sequential microcontroller implementation.

Advantages

- High-speed, deterministic operation due to true hardware parallelism, with PWM generation, echo timing and UART transmission running as independent concurrent processes.
- Modularity: each functional block (servo controller, ultrasonic sensor, UART TX, Radar UART) is implemented as an independently testable Verilog module.
- Low system cost, built from inexpensive, widely available components (HC-SR04, SG90, FTDI module).
- Real-time operation with no software-induced timing jitter, since all timing critical functions are implemented directly in hardware.

Limitations

- The HC-SR04 sensor has a relatively short effective range and limited angular beamwidth, restricting the radar's effective field of view.
- Measurement accuracy is sensitive to the surface properties of the target, including material, angle of incidence and surface texture, which can cause weak or absent echo returns.
- The system is currently restricted to a single forward sweep arc rather than full 360-degree coverage.
- The wired UART/FTDI link constrains the operating range between the FPGA hardware and the host PC visualization.

Applications

- Robotics: obstacle mapping and environment sensing for mobile robot navigation.
- Obstacle avoidance systems for autonomous vehicles and drones.
- Automation: proximity sensing in industrial and home-automation contexts.
- Education: a hands-on platform for teaching FPGA design, digital timing circuits and sensor interfacing concepts.

Future Scope

- Extending the sweep mechanism to achieve full 360-degree scanning coverage.
- Incorporating wireless communication (e.g., Bluetooth or Wi-Fi) to replace the wired UART/FTDI link to the host PC.
- Adding an onboard TFT display to enable standalone radar visualization without a host PC.
- Implementing object tracking algorithms to follow moving targets across successive sweep cycles.

Conclusion

This project has successfully demonstrated the design and implementation of an FPGA-based real-time ultrasonic radar system using an Artix-7 FPGA. By implementing servo control, ultrasonic echo timing and UART data transmission as independent, concurrently operating hardware modules in Verilog, the system achieves deterministic, jitter-free real-time performance that highlights the advantages of FPGA-based design over conventional microcontroller approaches for timing-critical embedded applications. The resulting radar visualization, rendered live in Processing 4, validates the correctness and synchronization of the complete hardware-to-host signal chain, and the project establishes a solid foundation for future enhancements including 360-degree scanning, wireless telemetry and object tracking.